

I. Introduction

I.1. What will we study?

Definition I.1.1 ((Algebraic) Number Field)

Let K be a subfield of $\mathbb{C}$ such that $\text{char}(K) = 0$. K is called a (algebraic) number field if

$$[K : \mathbb{Q}] < \infty$$

Example I.1.1

$$K = \mathbb{Q}, K = \mathbb{Q}[i], K = \mathbb{Q}[\sqrt{2}].$$

Definition I.1.2 (Algebraic Integer)

Let K be a number field. Say $\alpha \in K$ is an algebraic integer over K if α satisfies.

$$\alpha^n + a_1\alpha^{n-1} + \cdots + a_{n-1}\alpha + a_n = 0$$

where $a_1, a_2, \dots, a_n \in \mathbb{Z}$. That is, α is the root of some monic polynomial in $\mathbb{Z}[x]$.

We will basically study “Arithmetic”. E.g, what are all the homomorphisms from $K \rightarrow \mathbb{C}$? What is the “most interesting” subring of K ? It turns out that the following subring of K possesses interesting properties.

Definition I.1.3

Let $\mathcal{O}_K$ be the set

$$\mathcal{O}_K = \{\text{set of all algebraic integers in } K\}$$

which we call the ring of integers of a number field

This set has many interesting properties:

- $\mathcal{O}_K$ is a subring of K .
- Every element in K can be expressed as

$$\frac{a}{b} \quad a, b \in \mathcal{O}_K$$

- Every ideal of $\mathcal{O}_K$ can be uniquely expressed as product of prime ideals, up to reordering.
- The additive group of $\mathcal{O}_K$ is isomorphic to $\mathbb{Z}^n$.
- For every non-zero ideal I of $\mathcal{O}_K$, the additive group of I is a finite index abelian subgroup of $\mathcal{O}_K$.

In other words,

$$\mathcal{O}_K/I$$

is a finite abelian group.

II. Basic definitions in ring theory

Definition II.0.1

A ring R is a set together with two binary operations $+$ and $\times$ (called addition and multiplication) satisfying the following axioms:

- (1) $(R, +)$ is an abelian group.
- (2) $\times$ is associative.

(3) The distributive law holds.

We further impose the condition that

- (1) R is commutative.
- (2) R has a multiplicative identity.

Definition II.0.2

A subset I of R is called an ideal if

- I is a subring of R .
- For any $r \in R$, $rI \subset I$.

Definition II.0.3

Let I, J be two ideals in a ring R . Denote $I + J$ by

$$I + J = \{i + j \mid i \in I, j \in J\}$$

and IJ by

$$IJ = \left\{ \sum_{\text{finite}} i_k j_k \mid i_k \in I, j_k \in J \right\}$$

Definition II.0.4

An ideal P is called a prime ideal of ring R if

- $P \neq R$.
- For any $a, b \in R$ such that $ab \in P$, either $a \in P$ or $b \in P$.

Corollary II.0.1

I is a prime ideal of ring R if and only if R/I is an integral domain.